

GLOBAL FABRIC DAY 2026



Building Trusted Fabric Data Agent & Ontology



Wilson Mok

Jun 27, 2026

<https://globalfabric.community>

#GlobalFabricDay #GFD26



Wilson Mok

Director, Data and AI

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My experiences includes:

- Avanade
- CAE
- Air Canada

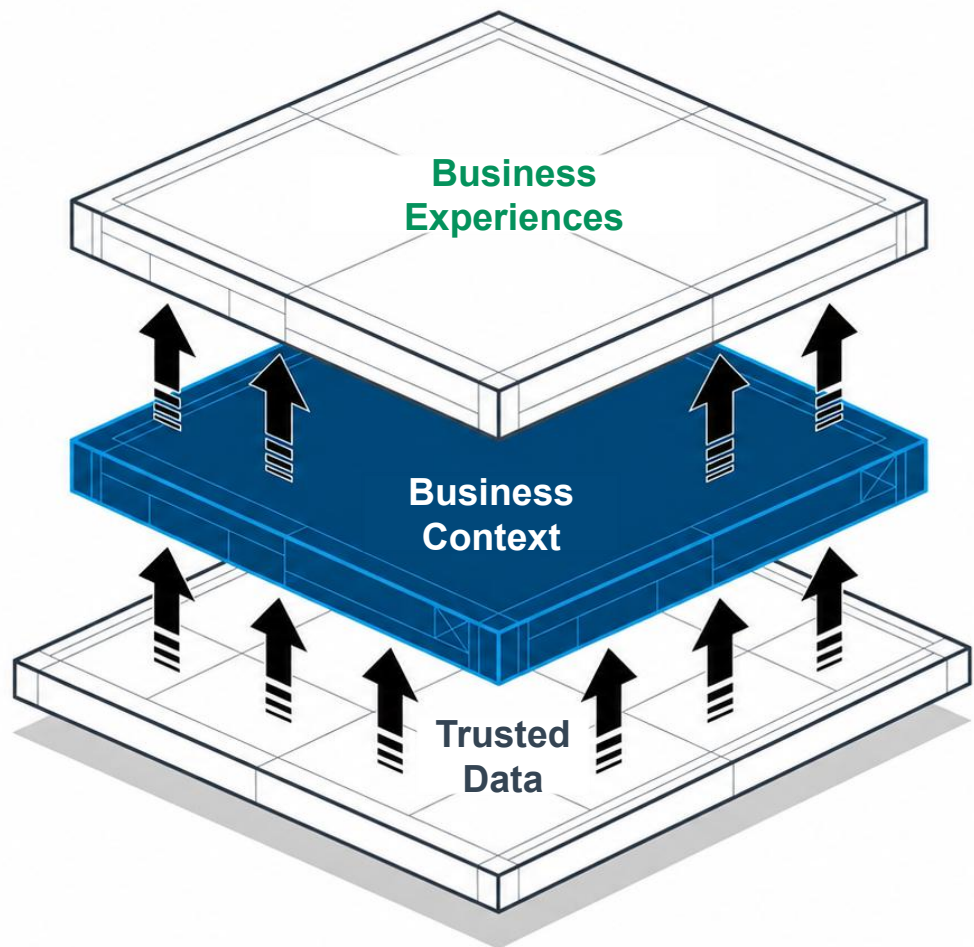


Expected Audience and Presentation Structure

- ✓ What We'll Cover:
 - Prepare your data for Fabric Data Agent.
 - How does Ontology and Graph improve business context?

⚙ Fundamental knowledge: Basic understanding Fabric and GenAI.

Trusted data prepares data for people. AI requires business context.



BUSINESS EXPERIENCES

Insights are consumed and actions are taken.

- Reports
- Fabric Data Agent
- Custom App (Rayfin)



BUSINESS CONTEXT

Provides the business meanings, relationships, and metadata that enable trusted AI experiences.

- Semantic Model
- Ontology
- Graph



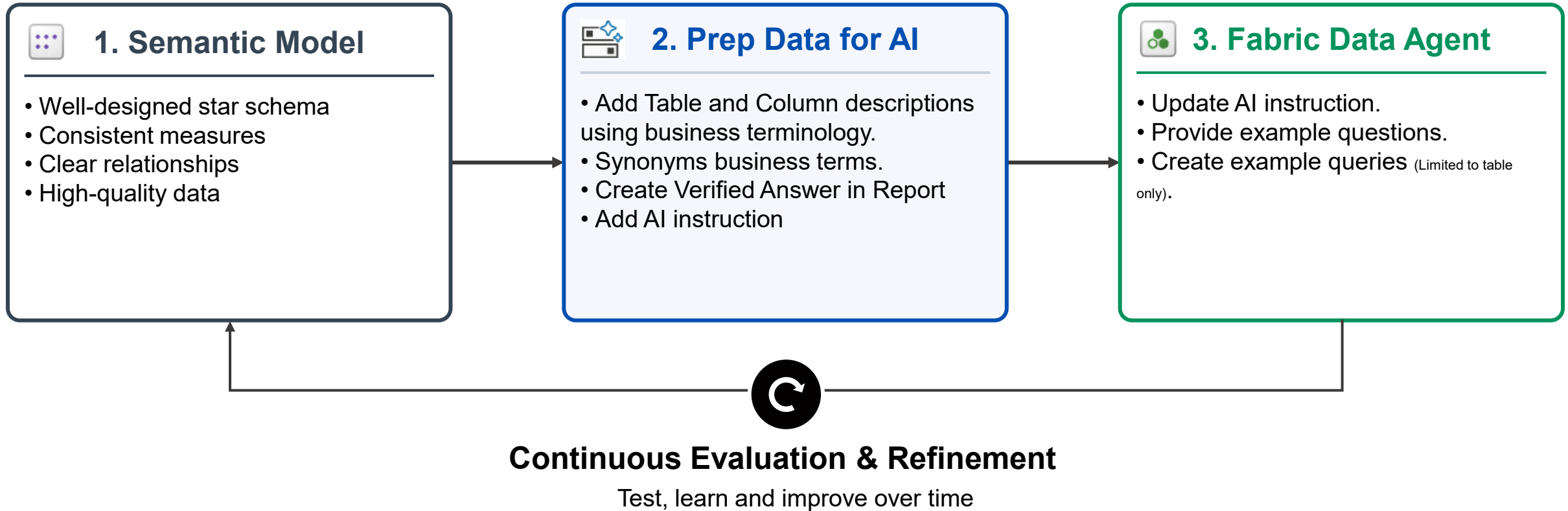
TRUSTED DATA

Delivers trusted, governed data ready for modeling and AI.

- Star Schema
- Data Warehouse
- Lakehouse
- Event House

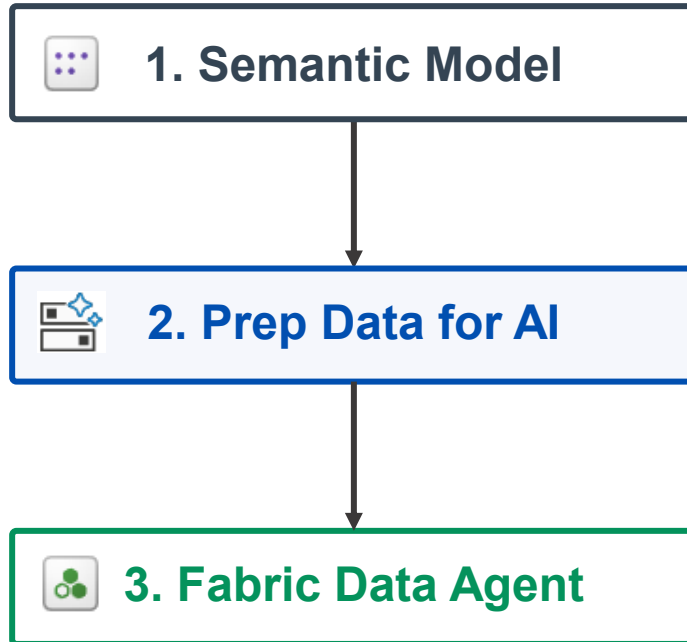
Key Takeaway: When context is added, AI-powered experiences become accurate, relevant and deliver values.

Preparing semantic model for AI extends beyond schema design.



Key Takeaway: Rather than embedding business context into every Data Agent, the Semantic Model centralizes metadata, relationships, and terminology, allowing multiple agents to deliver consistent, governed answers.

Demo 1: Prep Data for AI using Yelp Open Dataset.



Yelp Open Dataset

A real-world dataset designed for educational use.

yelp data licensing



6,990,280

Reviews



150,346

Businesses



11

Metropolitan areas



200,100

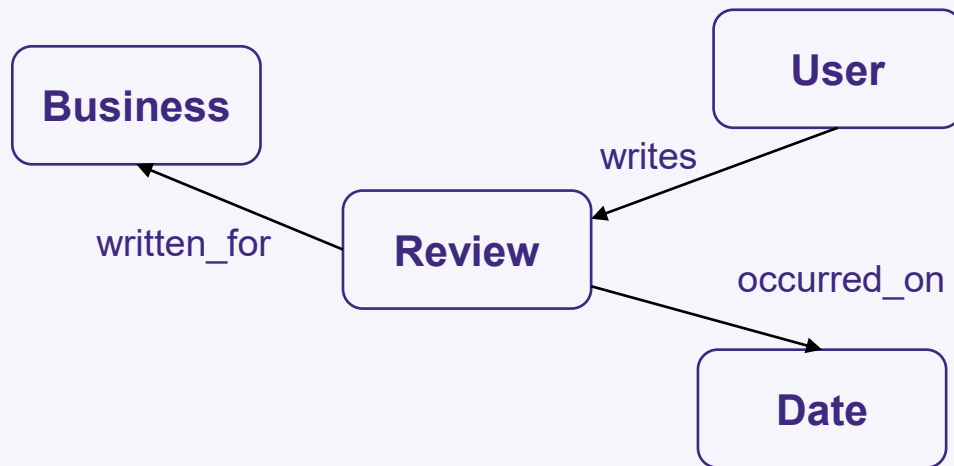
Pictures

Link: <https://business.yelp.com/data/resources/open-dataset/>

Looking ahead: Ontology and Graph take AI even further.

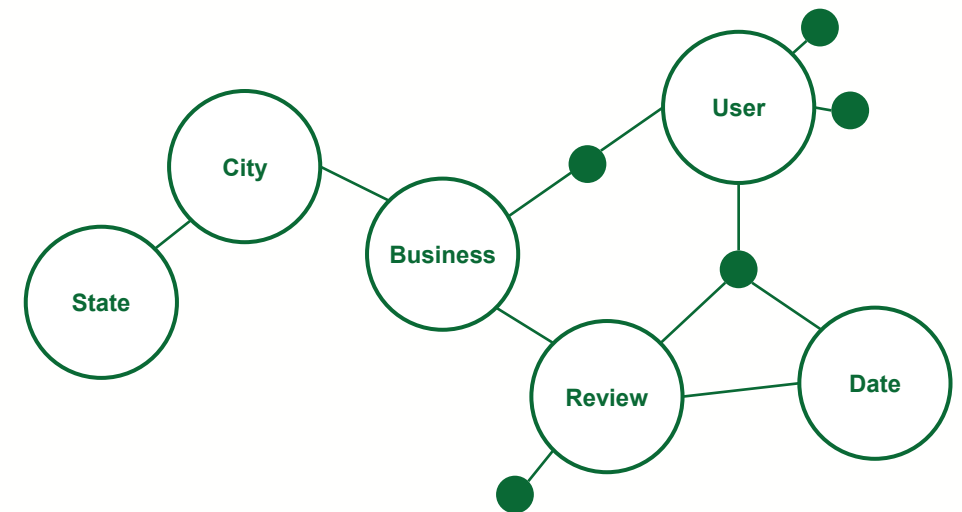
ONTOLOGY

- Captures the business meaning behind relationships, not just data connections.
- Defines relationship direction based on real business interactions rather than the underlying star schema.



GRAPH

- Connects the entities and relationships.
- Reveals hidden patterns and paths.
- Enables graph algorithms, dependency analysis, path finding.



Key Takeaway: As business context becomes richer, AI gains a deeper understanding of business concepts and relationships. Enabling more accurate reasoning and greater business impact.

Demo 2: Ontology and Graph take AI even further.

From Ontology to Graph:

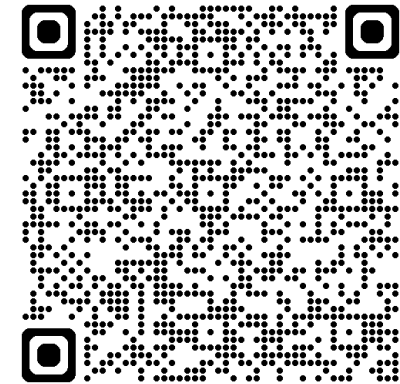
- How to generate an Ontology from Semantic model?
- Highlight the data profiling Ontology provides and the relationship graph.
- Run a Graph query to visualize how graph data is represented.
- Promote the City and State properties into a graph nodes to expose hidden relationships.

Key Takeaways: Business context is the bridge between trusted data and AI-powered business experiences.



- Semantic model is critical to the business context layer for Reporting and AI.
- The business context is built on metadata, business terminology and table relationships.
- Fabric Data Agents utilize all the context to deliver consistent, and accurate responds.
- Ontology and Graph represent the next evolution. It provides richer business concepts and relationships to enable more intelligent AI experiences.

Resources



GitHub Repo

<https://github.com/wilson-mok/demo/tree/main/2026/Global-Fabric-Day-MSDEVMTL/1-Build-Trusted-Fabric-Data-Agent-and-Ontology>

Thank you



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